SMIT — Batch 22

JavaScript Assignment 01: Fundamentals & Basic Problem Solving

Course:	Web & Mobile App Development
Subject:	JavaScript
Assignment:	01
Topics:	Chapters 1–9 — A Smarter Way to Learn JavaScript
Total Marks:	100
Duration:	5 Days
Submission:	GitHub Repository Link

Objective

The purpose of this assignment is to strengthen your understanding of JavaScript fundamentals, variables, data types, mathematical expressions, operators, and basic problem-solving skills.

You must write JavaScript code to solve each problem independently.

Instructions

1. Create a folder named `JS-Assignment-01`.
2. Create a separate JavaScript file for each question (`question-01.js`, etc.).
3. Use `prompt()` for user input wherever required.
4. Display results using `alert()`, `console.log()`, or `document.write()`.
5. Use meaningful variable names and proper JavaScript syntax.
6. Do not use loops, arrays, functions, or conditional statements in this assignment.
7. Do not use AI-generated solutions or copy code from classmates.
8. Push your completed assignment to a public GitHub repository.

Section A — Basic Understanding (20 Marks)

Question 1: Personal Information (5 Marks)

Create variables to store the following information:

- Full Name
- Age
- City
- Course Name
- Institute Name

Display all the information in a properly formatted introduction.

Expected Output:

```
My name is Ahmed.
I am 19 years old.
I live in Karachi.
I am learning Web & Mobile App Development.
My institute is SMIT.
```

Question 2: Variable Declaration (5 Marks)

Declare three variables using `var`, `let`, and `const`.

Store your favorite programming language, current learning topic, and batch number.

Display all three values in the console.

Question 3: Data Types (5 Marks)

Create variables containing the following data types:

- String
- Number
- Boolean
- Undefined
- Null

Use the `typeof` operator to display the data type of each variable.

Question 4: Legal and Illegal Variables (5 Marks)

Identify which variable names are valid or invalid. Write the reason for each invalid name.

```
var studentName;
var 1stStudent;
var _totalMarks;
var my-name;
var $price;
var class;
var userAge;
var first name;
var total_amount;
var let;
```

Correct all invalid variable names.

Section B — Mathematical Expressions (30 Marks)

Question 5: Basic Calculator (10 Marks)

Take two numbers from the user using `prompt()`.
Perform the following operations:

- Addition
- Subtraction
- Multiplication
- Division
- Modulus

Display the results.

Sample Input:

```
First Number: 20
Second Number: 6
```

Expected Output:

```
Addition: 26
Subtraction: 14
Multiplication: 120
Division: 3.3333333333333335
Modulus: 2
```

Question 6: Shopping Bill (10 Marks)

A customer purchases three products.
Ask the user to enter the price of each product and calculate:

1. Total bill
2. Discount amount (10%)
3. Final bill after discount

Sample Input:

```
Product 1: 1200
Product 2: 800
Product 3: 500
```

Expected Output:

```
Total Bill: Rs. 2500
Discount: Rs. 250
Final Bill: Rs. 2250
```

Question 7: Mathematical Expression (10 Marks)

Write JavaScript expressions to solve the following problems using the correct order of operations.

```
A = 10 + 5 * 2
B = (10 + 5) * 2
C = 20 / 2 + 3 * 4 - 5
D = 5 + 2 * (8 - 3) / 5
E = 30 % 4 + 6 * 2
```

Display each result and explain the calculation order using comments.

Section C — User Input & Strings (30 Marks)

Question 8: Welcome Message (10 Marks)

Ask the user for:

- First Name
- Last Name
- City

Combine the information and display a welcome message using string concatenation.

Sample Input:

```
First Name: Ali
Last Name: Khan
City: Karachi
```

Expected Output:

```
Welcome Ali Khan!
You are from Karachi.
```

Question 9: Age Calculator (10 Marks)

Ask the user for their birth year.
Calculate their approximate age using the current year.

Sample Input:

```
Birth Year: 2005
```

Expected Output (using 2026):

```
Your approximate age is 21 years.
```

Question 10: Marks Percentage Calculator (10 Marks)

Ask the user to enter marks for five subjects.
Each subject has a maximum of 100 marks.
Calculate:

1. Total obtained marks
2. Total maximum marks
3. Percentage

Sample Input:

```
English: 80
Math: 90
Science: 75
Computer: 95
Urdu: 85
```

Expected Output:

```
Obtained Marks: 425
Total Marks: 500
Percentage: 85%
```

Section D — Logic Challenge (20 Marks)

Question 11: Currency Converter (10 Marks)

Create a currency converter that converts Pakistani Rupees (PKR) into US Dollars (USD).
Use a fixed exchange rate:

```
1 USD = 280 PKR
```

Ask the user to enter an amount in PKR and display the equivalent amount in USD.

Sample Input:

```
Enter PKR: 5600
```

Expected Output:

```
Equivalent USD: $20
```

Question 12: Salary Calculator (10 Marks)

A company calculates employee salaries using the following structure:

- Basic Salary: Entered by the user
- House Allowance: 20% of basic salary
- Medical Allowance: 10% of basic salary
- Tax Deduction: 5% of basic salary

Calculate:

1. House allowance
2. Medical allowance
3. Tax deduction
4. Gross salary
5. Net salary after tax

Sample Input:

```
Basic Salary: 50000
```

Expected Output:

```
Basic Salary: Rs. 50000
House Allowance: Rs. 10000
Medical Allowance: Rs. 5000
Gross Salary: Rs. 65000
Tax Deduction: Rs. 2500
Net Salary: Rs. 62500
```

Submission Guidelines

GitHub Repository Name: `SMIT-Batch22-JS-Assignments`

Suggested folder structure:

```
SMIT-Batch22-JS-Assignments/
`-- Assignment-01/
    |-- index.html
    |-- question-01.js
    |-- question-02.js
    |-- ...
    |-- question-12.js
    `-- README.md
```

Students should submit:

- GitHub repository link
- All 12 completed questions
- README with student name, batch number, and assignment details
- Screenshots of outputs for at least three questions

Evaluation Criteria

Criteria	Marks
Correct Output	40
Logic & Calculations	30
Code Readability	15
GitHub Submission	10
Following Instructions	5
Total	**100**

Important Notes

- Every question must be completed.
- Use only the JavaScript concepts covered in class.
- Do not submit copied code.
- Be prepared to explain any question during class.
- Late submissions may be subject to marks deduction.

Best of luck, Batch 22!

Focus on understanding the logic, not just getting the correct output.